NEET UG STANDARD • PLANT PHYSIOLOGY

Photosynthesis in Higher Plants

Unit Context: Plant Physiology examines biological phenomena at the cellular and molecular levels while connecting them to organismic activities and environmental interactions. This tutorial covers the complete photosynthetic machinery, energy transformation mechanisms, carbon assimilation pathways, and limiting factors required for NEET UG mastery.

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1. Learning Objectives & Chapter Overview

All animals, including human beings, depend on plants for food. Green plants synthesise the organic food they need through **photosynthesis** and are therefore classified as **autotrophs**, whereas all other organisms that depend on green plants for nutrition are **heterotrophs**.

Core Definition & Ecological Significance

Photosynthesis is a physico-chemical process by which green plants use light energy to drive the synthesis of organic compounds. It is the fundamental basis of life on Earth for two primary reasons:

1. It is the **primary source of all food** on Earth.
2. It is responsible for the **release of molecular oxygen (O_2)** into the atmosphere.

Key Learning Objectives for NEET UG:

- Understand classical experiments establishing the roles of light, CO_2, chlorophyll, and H_2O in photosynthesis.
- Relate chloroplast ultrastructure to the division of labour between photochemical (light) and biosynthetic (dark) reactions.
- Analyze absorption and action spectra of photosynthetic pigments and the organization of Light Harvesting Complexes (PS I and PS II).
- Trace non-cyclic (Z-scheme) and cyclic electron transport, water splitting, and chemiosmotic ATP synthesis.
- Compare carbon fixation via the Calvin cycle (C_3), Hatch and Slack pathway (C_4), and photorespiration (C_2).
- Apply Blackman's Law of Limiting Factors to evaluate the effects of light, CO_2, temperature, and water on photosynthetic rates.

2. Foundational & Historical Experiments

2.1 Basic Classroom Demonstrations

- **Variegated Leaf / Black Paper Experiment:** Testing a variegated leaf or a leaf partially covered with black paper after exposure to light shows that starch formation occurs *only* in the green parts of the leaf in the presence of light.
- **Moll's Half-Leaf (KOH) Experiment:** Part of a leaf is enclosed in a test tube containing KOH-soaked cotton (which absorbs CO_2), while the other half remains exposed to air in light. Upon starch testing with iodine, the exposed half tests **positive** for starch, whereas the enclosed half inside the tube tests **negative**. This proves that CO_2 is essential for photosynthesis.

BOSONANCE

2.2 Milestone Historical Discoveries

Scientist & Period	Experimental Setup / Organism	Key Observation & Conclusion
Joseph Priestley (1733–1804) *Exp: 1770; Discovered O_2: 1774*	Closed bell jar containing a burning candle, a mouse, and a **mint plant**.	A burning candle extinguishes and a mouse suffocates in a closed bell jar because both "damage" the air. Placing a mint plant in the bell jar keeps the mouse alive and allows the candle to burn. **Hypothesis:** Plants restore to the air whatever breathing animals and burning candles remove.
Jan Ingenhousz (1730–1799)	Priestley's setup placed in dark vs. sunlight; aquatic plant (*Hydrilla*-type) in bright sunlight.	Showed that **sunlight is essential** for the plant process that purifies fouled air. In bright sunlight, small bubbles (O_2) formed around green parts of the aquatic plant, while none formed in the dark. Concluded that only **green parts** release oxygen.
Julius von Sachs (1854)	Plant growth and cellular localization studies.	Provided evidence that plants produce **glucose** when they grow, and glucose is usually stored as **starch**. Showed that the green substance (chlorophyll) is located in special bodies (later called **chloroplasts**) within plant cells.
T.W. Engelmann (1843–1909)	Prism-split light illuminating the green filamentous alga ***Cladophora*** placed in a suspension of **aerobic bacteria**.	Aerobic bacteria were used to detect sites of O_2 evolution. Bacteria accumulated mainly in the **blue and red light** regions of the split spectrum. Described the **first action spectrum** of photosynthesis, roughly resembling the absorption spectra of chlorophyll *a* and *b*.
Cornelius van Niel (1897–1985)	Purple and green sulphur bacteria.	Demonstrated that photosynthesis is a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduces CO_2 to carbohydrates. Proved that O_2 evolved by green plants comes from H_2O, not from CO_2.
Melvin Calvin (Born 1911; Nobel Prize 1961) & **J.A. Bassham**	Green algal photosynthesis using radioactive carbon-14 (^{14}C) labelled CO_2.	Proposed that plants convert light energy to chemical energy by transferring an electron in an organised array of pigment molecules. Mapped the complete pathway of carbon assimilation (**Calvin cycle**) and identified the first stable product as the 3-carbon acid **3-phosphoglyceric acid (PGA)**.



Tutorial Diagram 1: Joseph Priestley's Bell Jar Experiment (1770) demonstrating that plants restore air fouled by burning candles and breathing animals. (Note: To rekindle the candle without disturbing the closed setup, a converging lens/burning glass focusing sunlight can be used.)

2.3 Chemical Equations of Photosynthesis

By the mid-nineteenth century, the empirical equation representing oxygen-evolving photosynthesis was written as:

$$CO_2 + H_2O \xrightarrow{\text{Light}} [CH_2O] + O_2$$

where $[CH_2O]$ represents a carbohydrate (e.g., glucose, a 6-carbon sugar). Based on studies of purple and green sulphur bacteria, **Cornelius van Niel** formulated the generalised light-dependent reaction:

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$$2H_2A + CO_2 \xrightarrow{\text{Light}} 2A + CH_2O + H_2O$$

- In **green plants**, H_2O is the hydrogen donor and is oxidised to O_2.
- In **purple and green sulphur bacteria**, H_2S is the hydrogen donor, so the oxidation product is **sulphur or sulphate** (depending on the organism) and **not** O_2.
- This proved that O_2 evolved by green plants comes from H_2O, not from CO_2 (later confirmed using radioisotopic techniques with ^{18}O).

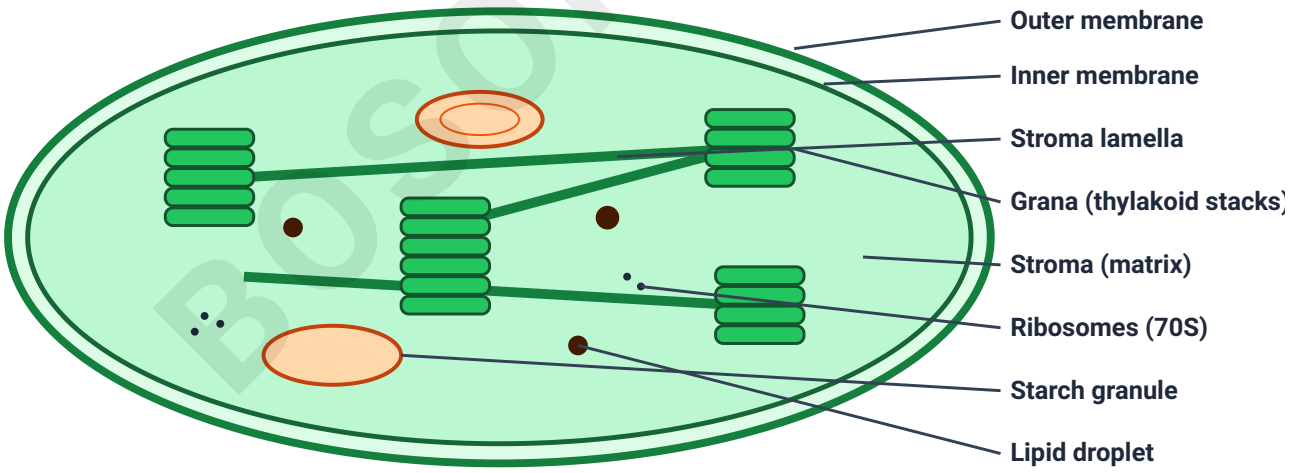
Because all $6\ O_2$ molecules (12 oxygen atoms) released per molecule of glucose synthesised must originate exclusively from water, **12 molecules of** H_2O are required as substrate on the left-hand side of the balanced multistep overall equation:

$$6CO_2 + 12H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

3. Site of Photosynthesis & Division of Labour

Photosynthesis occurs mainly in green leaves, as well as in other green parts of plants (such as green stems, unripe green fruits, and green sepals). Within leaves, **mesophyll cells** contain a large number of chloroplasts that align themselves along the cell walls to receive the optimum quantity of incident light:

- **Low / Optimum Light Intensity (Parastrophe):** Chloroplasts align with their flat surfaces **perpendicular** to the incident light (parallel to the periclinal/surface wall) to maximise light absorption.
- **Very High Light Intensity (Epistrophe):** Chloroplasts align with their flat surfaces **parallel** to the incident light (along the anticlinal/lateral walls) to protect chlorophyll from photo-oxidation.



Tutorial Diagram 2: Sectional view of a chloroplast showing the membranous system (grana and stroma lamellae) and the fluid stroma.

Division of Labour Within the Chloroplast

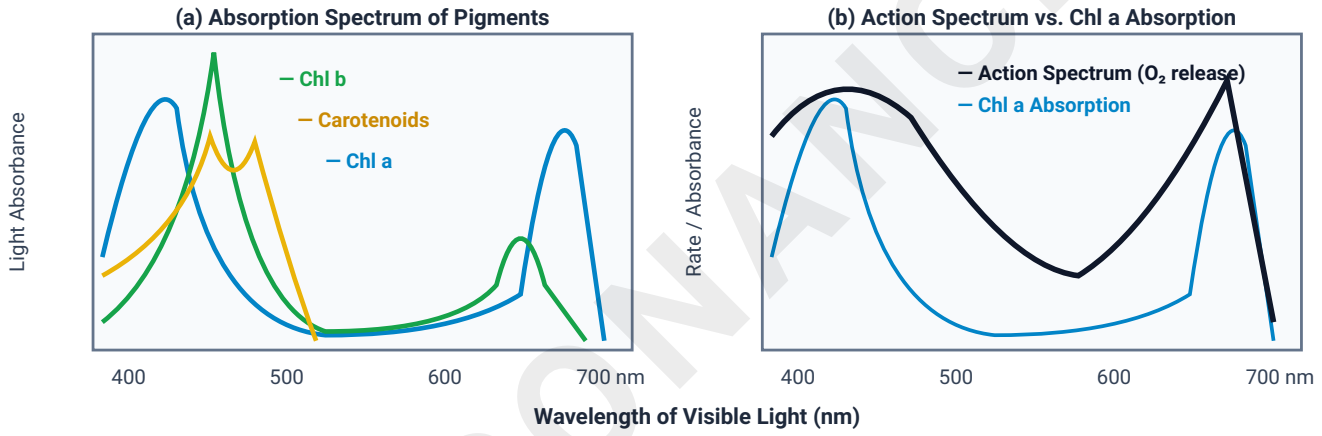
- **Membranous System (Grana + Stroma Lamellae):** Responsible for trapping light energy and synthesising **ATP and NADPH**. Because these reactions are directly light-driven, they are called **light reactions** (photochemical reactions).
- **Matrix Stroma:** Site of enzymatic reactions that synthesise sugar (which in turn forms starch). Because these reactions are not directly light-driven but depend on the products of light reactions (ATP and NADPH, besides CO_2 and H_2O), they are conventionally called **dark reactions** (carbon reactions /

biosynthetic phase). *Note: Calling them "dark reactions" is a misnomer because they do not require darkness and are indirectly light-dependent.*

4. Photosynthetic Pigments, Absorption & Action Spectra

Paper chromatographic separation of leaf pigments reveals that leaf colour is due to **four main pigments**:

Pigment Name	Chromatogram Colour	Functional Classification & Role
Chlorophyll *a*	Bright or blue-green	**Chief / Primary pigment** associated with photosynthesis; forms the reaction centre of PS I and PS II.
Chlorophyll *b*	Yellow-green	**Accessory pigment**; absorbs light and transfers energy to chlorophyll *a*.
Xanthophylls	Yellow	**Accessory pigment**; widens usable wavelength range and protects chlorophyll *a* from photo-oxidation.
Carotenoids	Yellow to yellow-orange	**Accessory pigment**; more stable than chlorophyll (leaves turn yellow in dark as chlorophyll degrades).



Tutorial Diagram 3: (a) Absorption spectra of chlorophyll a, chlorophyll b, and carotenoids; (b) Action spectrum of photosynthesis superimposed on the absorption spectrum of chlorophyll a.

Key Conclusions from Absorption & Action Spectra

- **Chlorophyll *a*** shows maximum absorption in the **blue** region (highest peak) and a second major absorption peak in the **red** region.
- Maximum rate of photosynthesis (measured by O_2 release) also occurs in the **blue and red regions** of the spectrum.
- There is **not a complete one-to-one overlap** between the absorption spectrum of chlorophyll *a* and the action spectrum of photosynthesis because **accessory pigments** (chlorophyll *b*, xanthophylls, carotenoids) absorb light at intermediate wavelengths and transfer the energy to chlorophyll *a*.
- **Functions of Accessory Pigments:** (1) Enable a wider range of incoming wavelengths to be utilised for photosynthesis, and (2) Protect chlorophyll *a* from **photo-oxidation**.

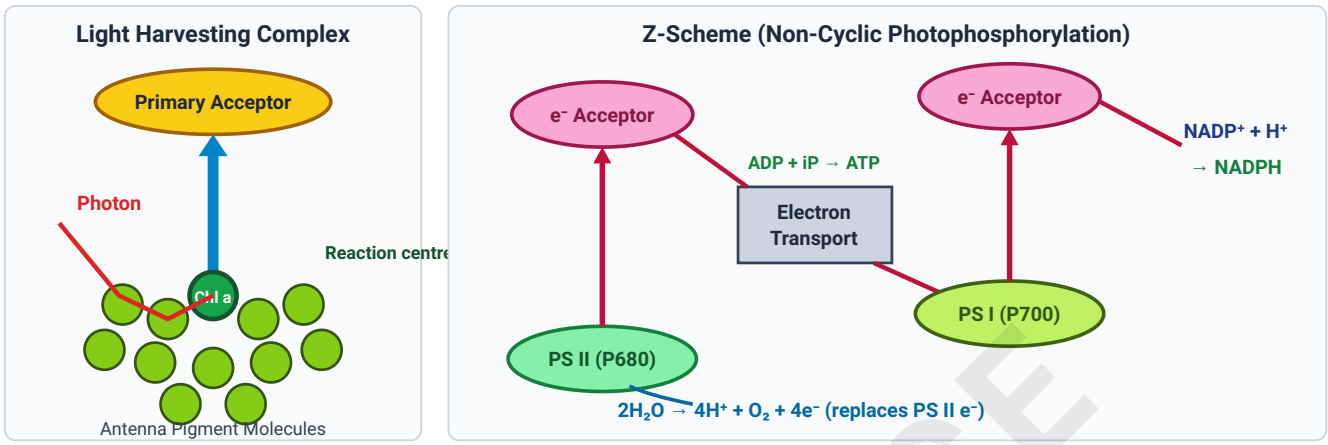
5. Light Reaction, Photophosphorylation & Chemiosmosis

5.1 Organization of Photosystems (LHC)

The photochemical phase includes: (1) light absorption, (2) water splitting, (3) oxygen release, and (4) formation of high-energy chemical intermediates (**ATP and NADPH**). Pigments are organised into two discrete photochemical **Light Harvesting Complexes (LHC)** within **Photosystem I (PS I)** and **Photosystem II (PS II)**. They are named in the sequence of their discovery, not in the sequence in which they function during the light reaction.

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- **Antennae (LHC):** Hundreds of pigment molecules bound to proteins (all pigments except one molecule of chlorophyll *a*) that absorb different wavelengths of light and funnel energy to the reaction centre.
- **Reaction Centre:** A single molecule of **chlorophyll *a***.
  - In **PS I**, the reaction centre chlorophyll *a* has an absorption peak at $700\ \text{nm}$ and is called **P700**.
  - In **PS II**, the reaction centre chlorophyll *a* has an absorption maximum at $680\ \text{nm}$ and is called **P680**.



Tutorial Diagram 4: Organization of the Light Harvesting Complex (left) and the Z-scheme of electron transport in non-cyclic photophosphorylation (right).

5.2 The Electron Transport (Z-Scheme) & Splitting of Water

1. **Excitation at PS II:** Reaction centre P680 absorbs $680\ \text{nm}$ red light, exciting electrons to jump to a farther atomic orbit.
2. **Uphill Transfer & Downhill ETS:** Excited electrons are picked up by a primary electron acceptor (uphill on the redox scale) and passed **downhill** along an oxidation-reduction (redox potential) scale through an Electron Transport System (ETS) consisting of **cytochromes** to PS I.
3. **Excitation at PS I & Reduction of NADP^+:** Simultaneously, electrons in P700 are excited by $700\ \text{nm}$ red light, transferred uphill to another acceptor with a greater redox potential, and then moved downhill to reduce energy-rich NADP^+ to $\text{NADPH} + \text{H}^+$.
4. **Splitting of Water (Photolysis):** Associated exclusively with **PS II** and physically located on the **inner side of the thylakoid membrane (facing the lumen)**. It continuously supplies electrons to replace those lost by PS II (which in turn replace those removed from PS I), while releasing protons (H^+) and O_2 into the **thylakoid lumen**:

$$2\text{H}_2\text{O} \longrightarrow 4\text{H}^+ + \text{O}_2 + 4\text{e}^-$$

5.3 Non-Cyclic vs. Cyclic Photophosphorylation

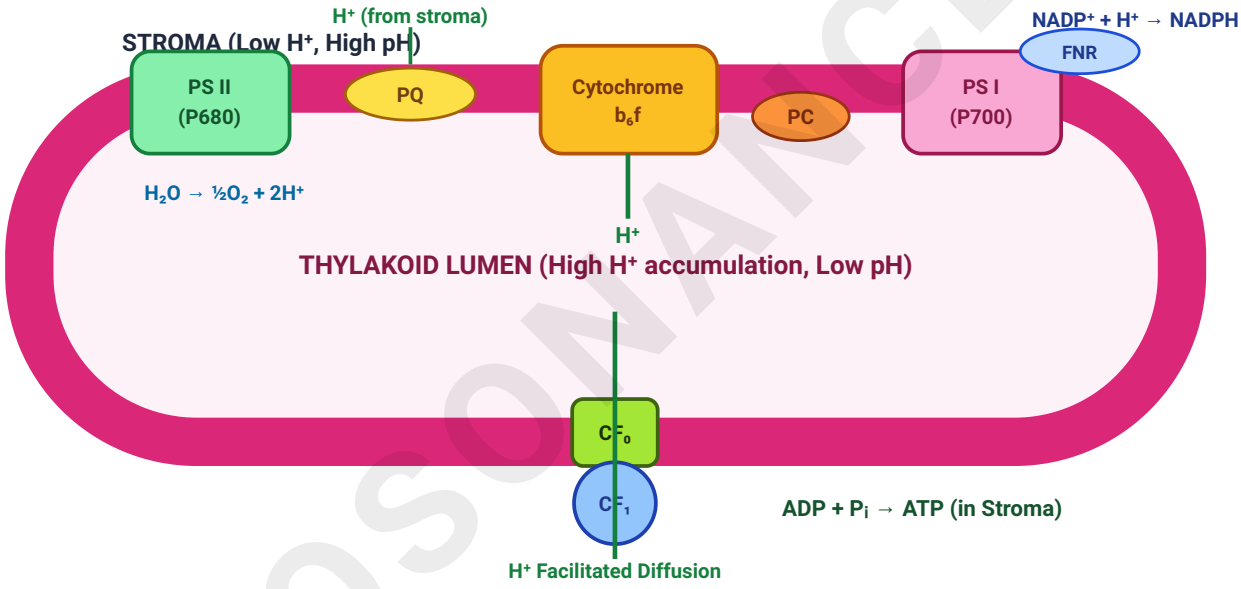
Photophosphorylation is the synthesis of ATP from ADP and inorganic phosphate (P_i) in the presence of light.

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Feature	Non-Cyclic Photophosphorylation	Cyclic Photophosphorylation
Photosystems Involved	Both **PS II and PS I** working in series (connected via ETS in the Z-scheme).	**Only PS I (P700)** is functional.
Location in Chloroplast	**Grana lamellae (thylakoids)**, which possess both PS I, PS II, and NADP reductase.	**Stroma lamellae**, which lack PS II and lack the NADP reductase enzyme.
Electron Flow Path	Unidirectional: $\mathrm{H_2O} \rightarrow \mathrm{PS\ II} \rightarrow \mathrm{ETS} \rightarrow \mathrm{PS\ I} \rightarrow \mathrm{NADP^+}$.	Cyclic: Excited $\mathrm{e^-}$ from P700 acceptor cycles back to PS I through the ETS.
Wavelength Requirement	Requires wavelengths supporting both $680\ \mathrm{nm}$ and $700\ \mathrm{nm}$.	Occurs when **only light of wavelengths beyond** $680\ \mathrm{nm}$ is available for excitation.
Products Synthesised	**Both** ATP **and** $\mathrm{NADPH + H^+}$ (plus $\mathrm{O_2}$ evolved from water splitting).	**Only** ATP (no $\mathrm{NADPH + H^+}$ synthesis; no $\mathrm{O_2}$ evolution).

5.4 Chemiosmotic Hypothesis of ATP Synthesis

ATP synthesis in chloroplasts is linked to the development of a **proton gradient** across the **thylakoid membrane**. Unlike respiration (where protons accumulate in the intermembrane space of mitochondria), in photosynthesis **proton accumulation occurs towards the inside of the membrane, i.e., in the thylakoid lumen**.



Tutorial Diagram 5: Mechanism of ATP synthesis through chemiosmosis across the thylakoid membrane.

Three Steps Causing the Proton Gradient Across the Thylakoid Membrane

1. **Water Splitting in the Lumen:** Splitting of water takes place on the inner side of the thylakoid membrane, so the $\mathrm{H^+}$ ions produced accumulate directly within the **lumen**.
2. **Proton Transport via Hydrogen Carrier (Plastoquinone):** As electrons move through the photosystems, the primary electron acceptor located towards the outer side of the membrane transfers its electron not to an electron carrier, but to an H**-carrier (plastoquinone)**. It removes a proton from the **stroma** while transporting an electron, and releases this proton into the **lumen** when passing its electron to the electron carrier (cytochrome b_6f) on the inner side of the membrane.
3. **Proton Removal by** NADP **Reductase in the Stroma:** The NADP reductase enzyme (FNR) is located on the **stroma side** of the membrane. Protons are removed from the stroma to reduce $\mathrm{NADP^+}$ to $\mathrm{NADPH + H^+}$.

Result: Protons decrease in number in the stroma and accumulate in the lumen, creating a strong proton gradient (electrochemical potential gradient) and a **measurable decrease in pH in the thylakoid lumen**.

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Requirements & Action of Chemiosmosis: Chemiosmosis requires (1) a **membrane**, (2) a **proton pump**, (3) a **proton gradient**, and (4) **ATP synthase**. The ATP synthase enzyme consists of two parts:

- CF_0**:** Embedded in the thylakoid membrane; forms a transmembrane channel that carries out **facilitated diffusion of protons** across the membrane from lumen to stroma.
- CF_1**:** Protrudes on the outer surface of the thylakoid membrane facing the **stroma**. Breakdown of the proton gradient releases energy that causes a **conformational change in the CF_1 particle**, activating the enzyme to synthesise ATP.

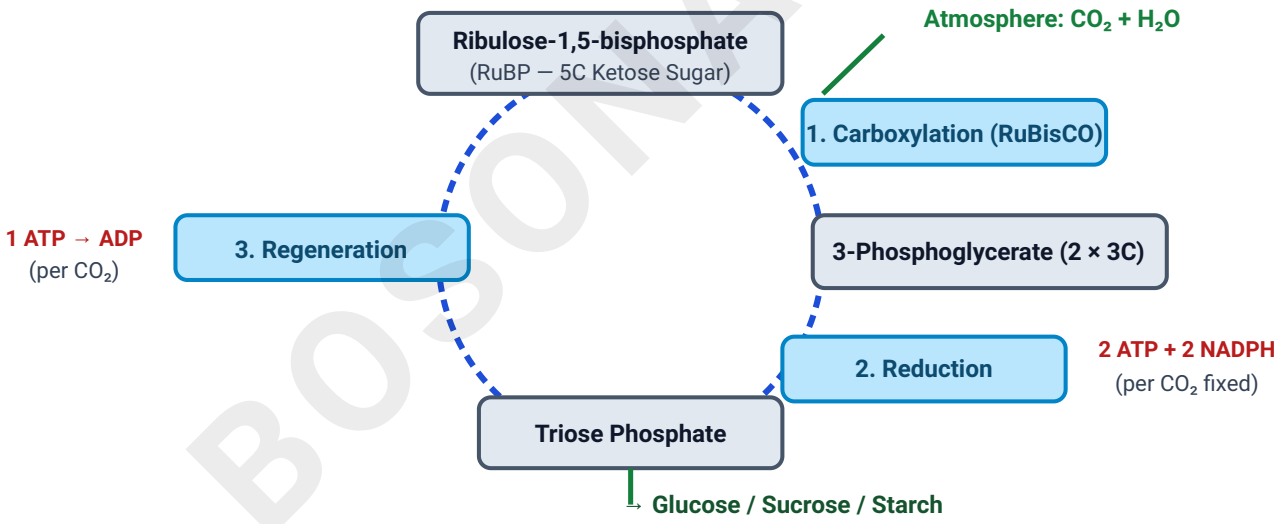
6. Biosynthetic Phase: Calvin Cycle (C_3), C_4 Pathway & Photorespiration

Of the light reaction products (ATP, $NADPH$, and O_2), O_2 diffuses out of the chloroplast while ATP and $NADPH$ drive the synthesis of sugars in the stroma. If light becomes unavailable, the biosynthetic process continues for a short time (until the accumulated ATP and $NADPH$ are exhausted) and then stops; it resumes when light is restored.

- C_3 **Pathway:** First stable product of CO_2 fixation is a 3-carbon organic acid, **3-phosphoglyceric acid (PGA)**.
- C_4 **Pathway:** First stable product of CO_2 fixation is a 4-carbon organic acid, **oxaloacetic acid (OAA)**.

6.1 The Calvin Cycle (C_3 Pathway)

Universal Occurrence: The Calvin cycle operates in a cyclic manner in **all photosynthetic plants**, regardless of whether they have C_3 or C_4 (or any other) pathways. The primary acceptor of CO_2 is a **5-carbon ketose sugar** called **ribulose-1,5-bisphosphate (RuBP)** (scientists initially searched in vain for a 2-carbon acceptor before discovering the 5-carbon RuBP).



Tutorial Diagram 6: The three stages of the Calvin cycle: (1) Carboxylation, (2) Reduction, and (3) Regeneration.

Three Stages of the Calvin Cycle:

1. **Carboxylation:** Fixation of CO_2 into a stable organic intermediate. This is the **most crucial step** of the Calvin cycle, where CO_2 carboxylates RuBP to form **two molecules of 3-PGA**, catalysed by **RuBP carboxylase-oxygenase (RuBisCO)**.
2. **Reduction:** A series of reactions leading to the formation of glucose. It utilises **2 molecules of ATP** (for phosphorylation) and **2 molecules of NADPH** (for reduction) **per CO_2 molecule fixed**.
3. **Regeneration:** Regeneration of the CO_2 acceptor molecule **RuBP** is crucial for the cycle to continue uninterrupted. It requires **1 molecule of ATP** for phosphorylation to form RuBP per CO_2 fixed.

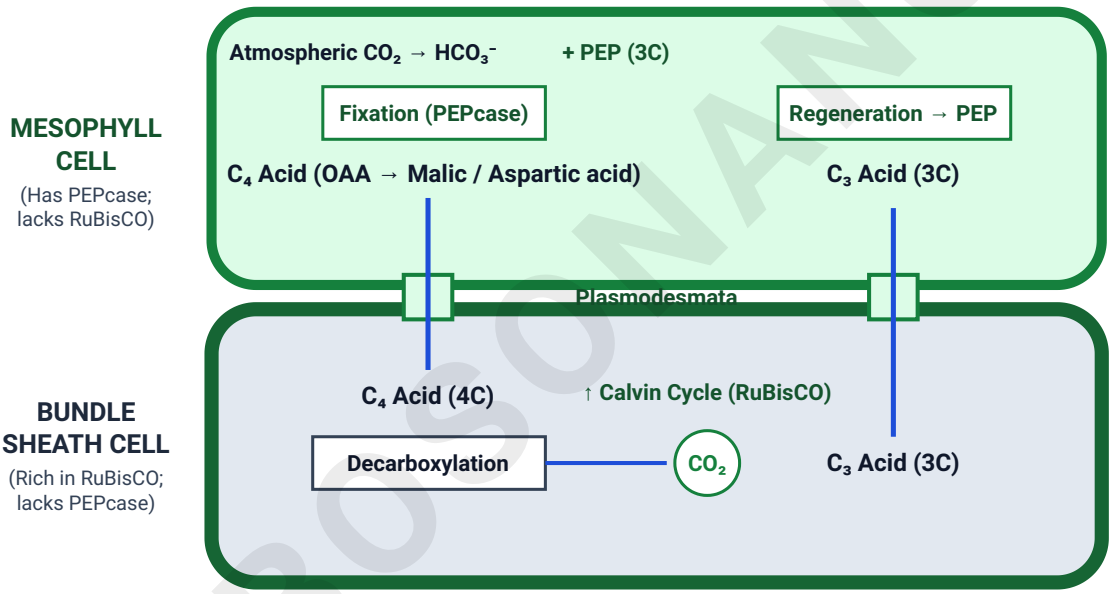
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Stoichiometric Basis	CO_2 Fixed	ATP Consumed	NADPH Consumed
Per 1 Turn of Calvin Cycle	$1\,CO_2$	$3\,ATP$ (2 in Reduction + 1 in Regeneration)	$2\,NADPH$ (in Reduction)
For 1 Glucose Molecule (6 Turns)	$6\,CO_2$	$18\,ATP \rightarrow 18\,ADP$	$12\,NADPH \rightarrow 12\,NADP^+$

6.2 The C_4 Pathway (Hatch and Slack Pathway)

Plants adapted to **dry tropical regions** (such as **maize** and **sorghum**) exhibit the C_4 pathway. Although oxaloacetic acid (4C) is their first CO_2 fixation product, they still use the **Calvin cycle (C_3 pathway)** as their main biosynthetic pathway to make sugars.

- **Special Characteristics of C_4 Plants:** Possess special leaf anatomy (**Kranz anatomy**), tolerate higher temperatures, show a response to high light intensities, **lack photorespiration**, and have greater productivity of biomass.
- **Kranz Anatomy ('Kranz' = 'wreath'):** Particularly large cells around the vascular bundles are called **bundle sheath cells**, which may form several layers around the vascular bundles. Bundle sheath cells are characterised by:
  1. A **large number of chloroplasts**,
  2. **Thick walls impervious to gaseous exchange**, and
  3. **No intercellular spaces**.



Tutorial Diagram 7: Diagrammatic representation of the Hatch and Slack (C_4) pathway operating across mesophyll and bundle sheath cells.

Step-by-Step Mechanism of the Hatch and Slack Pathway:

1. **Initial Fixation in Mesophyll Cells:** The primary CO_2 acceptor is a 3-carbon molecule, **phosphoenol pyruvate (PEP)**, present in mesophyll cells. Fixation is catalysed by **PEP carboxylase (PEPcase)** to form the 4-carbon **oxaloacetic acid (OAA)**. *Mesophyll cells of C_4 plants lack the RuBisCO enzyme.*
2. **Conversion & Transport:** OAA is converted into other 4-carbon compounds such as **malic acid or aspartic acid** in the mesophyll cells, which are then transported via plasmodesmata to the **bundle sheath cells**.
3. **Decarboxylation & Calvin Cycle in Bundle Sheath Cells:** In the bundle sheath cells, these C_4 acids are broken down (decarboxylated) to release CO_2 and a **3-carbon molecule**. The released CO_2 enters the **Calvin cycle**. *Bundle sheath cells are rich in RuBisCO but lack PEPcase.*
4. **Regeneration of PEP:** The 3-carbon molecule is transported back to the mesophyll cell, where it is converted back to **PEP** (consuming additional ATP), completing the cycle.

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6.3 Photorespiration (C_2 Pathway)

RuBisCO is the **most abundant enzyme in the world**. Its active site can bind to **both** CO_2 **and** O_2 competitively. RuBisCO has a much greater affinity for CO_2 than for O_2 when the $CO_2 : O_2$ ratio is nearly equal, and it is the **relative concentration of** O_2 **and** CO_2 that determines which gas binds to the enzyme:

- **Normal Carboxylation:** $\text{RuBP} + \text{CO}_2 \xrightarrow{\text{RuBisCO}} 2 \times \text{3-PGA (3C)}$.
- **Photorespiration (Oxygenation in C_3 Plants):** In C_3 plants, some O_2 binds to RuBisCO, decreasing CO_2 fixation. Instead of forming two molecules of 3-PGA, one molecule of RuBP (5C) binds with O_2 to form **one molecule of phosphoglycerate (3C)** and **one molecule of phosphoglycolate (2C)**:

$$\text{RuBP (5C)} + \text{O}_2 \xrightarrow{\text{RuBisCO (Oxygenase)}} 1\ \text{Phosphoglycerate (3C)} + 1\ \text{Phosphoglycolate (2C)}$$

- **Why Photorespiration is a Wasteful Process:** In the photorespiratory pathway, there is **neither synthesis of sugars nor synthesis of ATP or NADPH**. Instead, it results in the **release of** CO_2 **with the utilisation of ATP**. Its biological function is not yet known.
- **Why C_4 Plants Lack Photorespiration:** C_4 plants have a mechanism that increases the intracellular concentration of CO_2 at the RuBisCO enzyme site when C_4 acids from the mesophyll are broken down in the bundle sheath cells to release CO_2 (and thick bundle sheath walls prevent gas leakage). This ensures that RuBisCO functions almost exclusively as a **carboxylase**, minimising its oxygenase activity.

6.4 Complete Comparison Between C_3 and C_4 Plants

Characteristic	C_3 Plants	C_4 Plants
Cell type in which the Calvin cycle takes place	**Mesophyll**	**Bundle sheath**
Cell type in which initial carboxylation reaction occurs	**Mesophyll**	**Mesophyll**
How many cell types does the leaf have that fix CO_2?	**One:** Mesophyll	**Two:** Bundle sheath and mesophyll
Which is the primary CO_2 acceptor?	**RuBP** (Ribulose-1,5-bisphosphate)	**PEP** (Phosphoenol pyruvate)
Number of carbons in the primary CO_2 acceptor	**5**	**3**
Which is the primary CO_2 fixation product?	**PGA** (3-phosphoglyceric acid)	**OAA** (Oxaloacetic acid)
Number of carbons in the primary CO_2 fixation product	**3**	**4**
Does the plant have RuBisCO?	**Yes**	**Yes**
Does the plant have PEPcase?	**No** (lacks C_4 photosynthetic PEPcase)	**Yes** (in mesophyll cells)
Which cells in the plant have RuBisCO?	**Mesophyll**	**Bundle sheath**
CO_2 fixation rate under high light conditions	**Medium / Low** (limited by CO_2 and photorespiration)	**High**
Whether photorespiration is present at low light intensities	**Negligible**	**Negligible**
Whether photorespiration is present at high light intensities	**High**	**Negligible**
Whether photorespiration is present at low CO_2 concentrations	**High**	**Negligible**
Whether photorespiration is present at high CO_2 concentrations	**Negligible**	**Negligible**
Temperature optimum	**20–25 °C**	**30–40 °C**
Representative Examples	Tomato, Bell pepper, Wheat, Rice	Maize, Sorghum, Sugarcane

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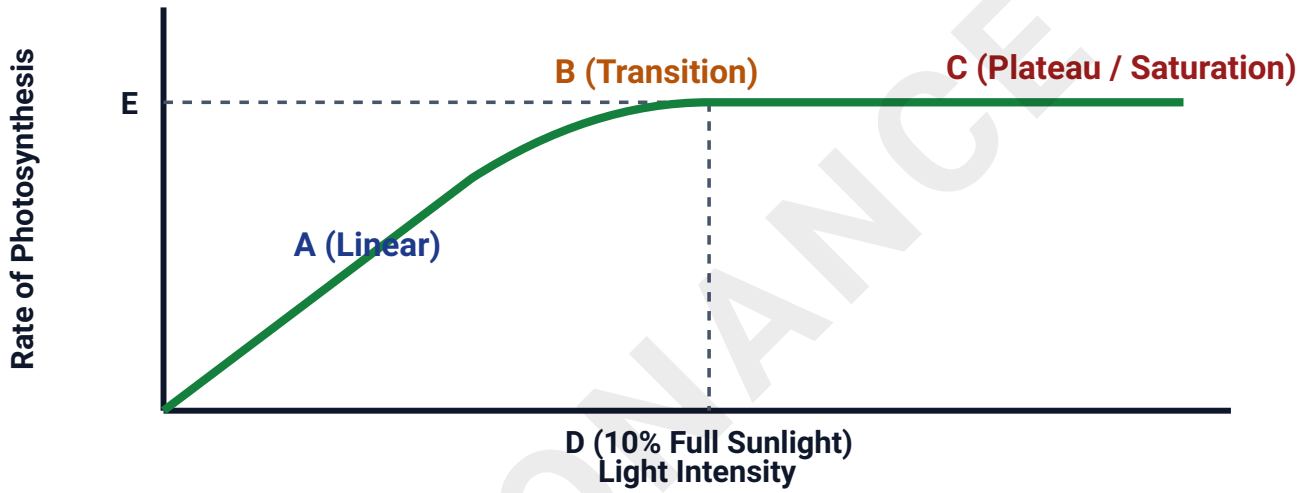
7. Factors Affecting Photosynthesis

- **Internal (Plant) Factors:** Number, size, age, and orientation of leaves, mesophyll cells, and chloroplasts; internal CO_2 concentration; and the amount of chlorophyll. These depend on the **genetic predisposition** and the growth of the plant.
- **External Factors:** Availability of sunlight, temperature, CO_2 concentration, and water.

Blackman's Law of Limiting Factors (1905)

"If a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value: it is the factor which directly affects the process if its quantity is changed."

For example, a green leaf with optimal light and CO_2 will not photosynthesise if the temperature is very low; raising the temperature to optimum immediately allows photosynthesis to proceed.



Tutorial Diagram 8: Effect of light intensity on the rate of photosynthesis. Region A shows a linear relationship where light is limiting; Region B is the transition zone; Region C represents maximum rate (E) where other factors become limiting; Point D is the light saturation point (10% of full sunlight).

Detailed Analysis of External Factors

1. **Light (Quality, Intensity, and Duration):**
   - At low light intensities (Region A), there is a **linear relationship** between incident light and CO_2 fixation rates (light is the limiting factor).
   - At higher light intensities (Region C), the rate shows no further increase because other factors (CO_2, temperature) become limiting.
   - **Light saturation (Point D) occurs at 10% of full sunlight.** Consequently, except for plants in shade or dense forests, light is rarely a limiting factor in nature.
   - Increase in incident light beyond a point causes the **breakdown of chlorophyll (solarisation/photo-oxidation)** and a decrease in photosynthesis.

2. **Carbon Dioxide Concentration:**
   - CO_2 is the **major limiting factor** for photosynthesis. Atmospheric CO_2 concentration is very low (between **0.03% and 0.04%**, i.e., $300–400\ \mu L\ L^{-1}$).
   - Increasing concentration up to **0.05%** increases CO_2 fixation rates; beyond this level, it can become damaging over longer periods.
   - At low light conditions, neither C_3 nor C_4 plants respond to high CO_2 conditions.
   - At high light intensities, both show increased rates, but C_4 **plants show saturation at about** $360\ \mu L\ L^{-1}$, whereas C_3 **plants respond to increased** CO_2 **and show saturation only beyond** $450\ \mu L\ L^{-1}$. Thus,

current atmospheric CO_2 availability is limiting to C_3 plants.
- **Greenhouse Application:** C_3 crops such as **tomatoes and bell pepper** are grown in a CO_2-enriched atmosphere to achieve higher productivity and yields.

3. **Temperature:**
   - The **dark reactions, being enzymatic, are temperature-controlled**. Light reactions are also temperature-sensitive, but are affected to a much lesser extent.
   - C_4 plants respond to higher temperatures ($30–40\ ^\circ C$) and show higher rates of photosynthesis, while C_3 plants have a much lower temperature optimum ($20–25\ ^\circ C$).
   - Tropical plants have a higher temperature optimum than plants adapted to temperate climates.

4. **Water:**
   - Although water is a reactant in the light reaction, the effect of water as a limiting factor is **more through its effect on the plant as a whole** rather than directly on the photosynthetic reaction.
   - **Water stress** causes: (a) **stomata to close**, reducing CO_2 availability, and (b) **leaves to wilt**, reducing leaf surface area and overall metabolic activity.

8. Worked Examples & Conceptual Applications

Worked Example 1: External vs. Internal Identification of C_3 and C_4 Plants

Problem: Can you tell whether a plant is C_3 or C_4 merely by looking at it externally? Which internal structure definitively distinguishes them?

Solution:

- **External Observation:** Generally, one cannot conclusively tell whether a plant is C_3 or C_4 purely by external morphology, although C_4 plants are well adapted to dry tropical climates and do not show wilting/photorespiratory stress under high temperatures and bright sunlight.
- **Internal Anatomical Structure:** A vertical section of the leaf under a microscope definitively distinguishes them by the presence of **Kranz anatomy** in C_4 plants. C_4 leaves possess large **bundle sheath cells** forming wreath-like layers around the vascular bundles, with a large number of chloroplasts, thick walls impervious to gaseous exchange, and no intercellular spaces. C_3 leaves lack Kranz anatomy.

Worked Example 2: High Productivity in C_4 Plants Despite Few Calvin-Cycle Cells

Problem: Even though very few cells (only the bundle sheath cells) in a C_4 plant carry out the biosynthetic Calvin pathway, why are C_4 plants highly productive?

Solution: In C_4 plants, mesophyll cells rapidly fix CO_2 via PEPcase into C_4 acids, which are pumped into the bundle sheath cells and decarboxylated. This acts as an intracellular CO_2 pump that elevates the CO_2 concentration at the RuBisCO site inside bundle sheath cells. Consequently, RuBisCO acts exclusively as a carboxylase and **photorespiration is completely eliminated**, resulting in high photosynthetic rates and biomass yield.

Worked Example 3: Pigment Stability, Shade Adaptation, and Accessory Pigments

Problem: (a) Would a plant with a high concentration of chlorophyll *b* but lacking chlorophyll *a* carry out photosynthesis? (b) Why does a leaf kept in the dark turn yellow or pale green, and which pigment is more stable? (c) Why are leaves kept in the shade darker green than leaves on the sunny side?

Solution:

- **(a) No.** Chlorophyll *a* forms the **reaction centre** (P680 and P700) where light energy is converted into chemical energy via electron excitation. Without chlorophyll *a*, photosynthesis cannot occur. Chlorophyll

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b and other accessory pigments are present to harvest a wider range of wavelengths, transfer energy to chlorophyll *a*, and protect chlorophyll *a* from photo-oxidation.

- **(b)** Chlorophyll synthesis requires light, and existing chlorophyll degrades in the dark, unmasking the yellow colour of **carotenoids and xanthophylls**, which are **more stable** pigments.
- **(c)** Leaves in the **shade are darker green** because they synthesise a higher concentration of chlorophyll pigments (and orient their chloroplasts flat/perpendicular to incident rays) to maximise capture of the limited available light.

Worked Example 4: Bioenergetics Calculation for Hexose Synthesis

Problem: Calculate the total number of ATP and NADPH molecules required in the stroma to synthesise 2 molecules of glucose in a $\mathrm{C_3}$ plant, and specify how many ATP molecules are consumed solely in the regeneration stage.

Solution:

- For 1 molecule of glucose ($6\,\mathrm{CO_2}$ fixed = 6 turns of Calvin cycle):
  - Reduction stage uses $2\,\mathrm{ATP}$ and $2\,\mathrm{NADPH}$ per $\mathrm{CO_2} \Rightarrow 12\,\mathrm{ATP}$ and $12\,\mathrm{NADPH}$.
  - Regeneration stage uses $1\,\mathrm{ATP}$ per $\mathrm{CO_2} \Rightarrow 6\,\mathrm{ATP}$.
  - Total per glucose = $18\,\mathrm{ATP}$ and $12\,\mathrm{NADPH}$.
- For 2 molecules of glucose: Total required = $2 \times 18 = \mathbf{36}\,\mathrm{ATP}$ and $2 \times 12 = \mathbf{24}\,\mathrm{NADPH}$, of which $2 \times 6 = \mathbf{12}\,\mathrm{ATP}$ are used in the regeneration of RuBP.

9. NEET UG Practice Questions with Worked Solutions

Note: All questions below are original practice items authored strictly to the NEET UG examination standard to test conceptual clarity, multi-statement analysis,Assertion–Reason logic, and stoichiometric application.

Practice Question 1 (Single-Choice MCQ — Historical Experiments)

Which of the following correctly pairs the scientist with his experimental organism/setup and key conclusion regarding photosynthesis?

(1) Jan Ingenhousz — *Cladophora* and aerobic bacteria — First action spectrum of photosynthesis

(2) T.W. Engelmann — Purple and green sulphur bacteria — Oxygen evolved comes from water

(3) Cornelius van Niel — Purple and green sulphur bacteria — Photosynthesis is a light-dependent reaction where hydrogen from an oxidisable compound reduces $\mathrm{CO_2}$

(4) Julius von Sachs — Mint plant in a closed bell jar — Green plants restore air fouled by breathing animals

Correct Answer: (3)

Worked Solution: Cornelius van Niel (1897–1985) studied purple and green sulphur bacteria (which use $\mathrm{H_2S}$ as the hydrogen donor and release sulphur or sulphate instead of $\mathrm{O_2}$) and demonstrated that hydrogen from a suitable oxidisable compound reduces $\mathrm{CO_2}$ to carbohydrates in the presence of light. Engelmann used *Cladophora* and aerobic bacteria; Priestley used the mint plant and bell jar; and Sachs demonstrated glucose synthesis in chloroplasts and its storage as starch.

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Practice Question 2 (Single-Choice MCQ — Pigments & Chromatogram)

When leaf pigments of a green plant are separated by paper chromatography, the correct match between the pigment and its characteristic colour on the chromatogram is:

(1) Chlorophyll *a*: Yellow-green; Chlorophyll *b*: Bright or blue-green

(2) Chlorophyll *a*: Bright or blue-green; Xanthophylls: Yellow

(3) Carotenoids: Bright green; Xanthophylls: Yellow-orange

(4) Chlorophyll *b*: Yellow; Carotenoids: Blue-green

Correct Answer: (2)
Worked Solution: Chromatographic separation of leaf pigments yields four distinct pigment bands: **Chlorophyll *a*** appears **bright or blue-green**, **chlorophyll *b*** appears **yellow-green**, **xanthophylls** appear **yellow**, and **carotenoids** appear **yellow to yellow-orange**.

Practice Question 3 (Assertion–Reason — Photophosphorylation)

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Cyclic photophosphorylation operating in the stroma lamellae results only in the synthesis of ATP and not of $\mathrm{NADPH} + \mathrm{H}^{+}$.
Reason (R): The stroma lamellae membranes lack both Photosystem II and the NADP reductase enzyme.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)
Worked Solution: Unlike grana lamellae (which have both PS I and PS II), the stroma lamellae membranes lack PS II as well as the NADP reductase enzyme. Therefore, excited electrons from PS I (P700) cannot be passed on to NADP^{+} and are instead cycled back to the PS I complex through the electron transport chain, generating only ATP. Thus, both (A) and (R) are true, and (R) is the exact structural explanation of (A).

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Practice Question 4 (Multiple-Statement Evaluation — Chemiosmotic Hypothesis)

Read the following statements regarding the chemiosmotic synthesis of ATP in chloroplasts:
I. Proton accumulation takes place in the intermembrane space of the chloroplast.
II. The water-splitting complex is associated with PS II on the inner side of the thylakoid membrane.
III. The NADP reductase enzyme is located on the lumen side of the thylakoid membrane.
IV. Breakdown of the proton gradient across the membrane through the transmembrane channel of $\mathrm{CF_0}$ causes a conformational change in the $\mathrm{CF_1}$ particle to synthesise ATP.
Which of the above statements are **correct**?

(1) I and III only

(2) II and IV only

(3) II, III and IV only

(4) I, II and IV only

Correct Answer: (2)
Worked Solution: Statement I is incorrect because in chloroplasts, protons accumulate in the **thylakoid lumen** (accumulation in the intermembrane space occurs in mitochondria during respiration). Statement II is correct. Statement III is incorrect because the NADP reductase enzyme is located on the **stroma side** (outer surface) of the thylakoid membrane. Statement IV is correct. Hence, only II and IV are correct.

Practice Question 5 (Single-Choice MCQ — Water Splitting & Z-Scheme)

During non-cyclic electron transport (Z-scheme), the electrons released by the splitting of water molecules directly replace the electrons lost from:

(1) Reaction centre P700 of Photosystem I

(2) Reaction centre P680 of Photosystem II

(3) Primary electron acceptor of Photosystem II

(4) Ferredoxin and NADP reductase complex

Correct Answer: (2)
Worked Solution: Splitting of water ($\mathrm{2H_2O \rightarrow 4H^+ + O_2 + 4e^-}$) is associated with PS II. The electrons made available by splitting water directly replace the electrons moved from **Photosystem II (P680)**, while the electrons needed to replace those removed from Photosystem I (P700) are provided by Photosystem II via the electron transport chain.

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Practice Question 6 (Single-Choice MCQ — Calvin Cycle Energetics)

In the Calvin cycle, for every 3 molecules of $\mathrm{CO_2}$ fixed, how many molecules of ATP and NADPH are utilised respectively during the **reduction step alone**, and how many ATP molecules are utilised during the **regeneration step**?

(1) Reduction: 9 ATP, 6 NADPH; Regeneration: 3 ATP

(2) Reduction: 6 ATP, 6 NADPH; Regeneration: 3 ATP

(3) Reduction: 6 ATP, 3 NADPH; Regeneration: 6 ATP

(4) Reduction: 18 ATP, 12 NADPH; Regeneration: 6 ATP

Correct Answer: (2)
Worked Solution: Per molecule of $\mathrm{CO_2}$ fixed in the Calvin cycle, the **reduction** step utilises $2\,\mathrm{ATP}$ and $2\,\mathrm{NADPH}$, and the **regeneration** step utilises $1\,\mathrm{ATP}$. Therefore, for $3\,\mathrm{CO_2}$ molecules fixed: Reduction uses $3 \times 2 = \mathbf{6}\,\mathrm{ATP}$ and $3 \times 2 = \mathbf{6}\,\mathrm{NADPH}$, while Regeneration uses $3 \times 1 = \mathbf{3}\,\mathrm{ATP}$ (total $9\,\mathrm{ATP}$ and $6\,\mathrm{NADPH}$).

Practice Question 7 (Statement I / Statement II — $\mathrm{C_4}$ Pathway & Kranz Anatomy)

Given below are two statements:
Statement I: In $\mathrm{C_4}$ plants, mesophyll cells possess PEPcase for initial $\mathrm{CO_2}$ fixation but lack the RuBisCO enzyme, whereas bundle sheath cells are rich in RuBisCO but lack PEPcase.
Statement II: The Calvin pathway takes place in the mesophyll cells of $\mathrm{C_3}$ plants, whereas in $\mathrm{C_4}$ plants it takes place in both mesophyll and bundle sheath cells.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

Correct Answer: (3)
Worked Solution: Statement I is completely correct. Statement II is incorrect because in $\mathrm{C_4}$ plants, the Calvin cycle **does not take place in the mesophyll cells** (since they lack RuBisCO); it takes place **only in the bundle sheath cells**.

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Practice Question 8 (Single-Choice MCQ — Photorespiration)

During photorespiration in C_3 plants, when O_2 binds to the active site of RuBisCO instead of CO_2, one molecule of RuBP is converted into:

(1) Two molecules of 3-phosphoglyceric acid (3C)

(2) Two molecules of phosphoglycolate (2C)

(3) One molecule of phosphoglycerate (3C) and one molecule of phosphoglycolate (2C)

(4) One molecule of oxaloacetic acid (4C) and one molecule of CO_2

Correct Answer: (3)
Worked Solution: When O_2 competitively binds to RuBisCO in C_3 plants, the 5-carbon RuBP is oxygenated into **one molecule of 3-carbon phosphoglycerate** and **one molecule of 2-carbon phosphoglycolate**. The pathway consumes ATP and releases CO_2 without synthesising ATP, NADPH, or sugars.

Practice Question 9 (Matching Type — Enzymes, Substrates & Locations)

Match Column I with Column II and select the correct option:

Column I	Column II
A. Primary CO_2 acceptor in C_3 pathway	i. 3-carbon molecule (PEP)
B. Primary CO_2 acceptor in C_4 pathway	ii. 4-carbon organic acid (OAA)
C. First stable product of Hatch and Slack pathway	iii. 5-carbon ketose sugar (RuBP)
D. Reaction centre of Photosystem II	iv. Chlorophyll *a* with absorption maxima at $680\ nm$

(1) A–iii, B–i, C–ii, D–iv

(2) A–i, B–iii, C–ii, D–iv

(3) A–iii, B–ii, C–i, D–iv

(4) A–ii, B–i, C–iii, D–iv

Correct Answer: (1)
Worked Solution:

- **A–iii:** Primary CO_2 acceptor in the C_3 pathway is the 5-carbon ketose sugar ribulose bisphosphate (RuBP).
- **B–i:** Primary CO_2 acceptor in the C_4 pathway is the 3-carbon molecule phosphoenol pyruvate (PEP).
- **C–ii:** First stable product of the C_4 (Hatch and Slack) pathway is the 4-carbon oxaloacetic acid (OAA).
- **D–iv:** Reaction centre of PS II is P680 (chlorophyll *a* absorbing at $680\ nm$).

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Practice Question 10 (Single-Choice MCQ — Limiting Factors & CO_2 Saturation)

Which of the following statements accurately describes the response of C_3 and C_4 plants to light intensity and carbon dioxide concentration?

(1) Light saturation in nature occurs at 50% of full sunlight, making light the most frequent limiting factor for open-field crops.

(2) At low light conditions, C_3 plants respond strongly to elevated CO_2 concentrations whereas C_4 plants do not.

(3) At high light intensities, C_4 plants show CO_2 saturation at about $360\ \mu L\ L^{-1}$, while C_3 plants show saturation only beyond $450\ \mu L\ L^{-1}$.

(4) C_4 plants such as tomatoes and bell pepper are commercially grown in CO_2-enriched greenhouses to overcome CO_2 limitation.

Correct Answer: (3)

Worked Solution: Light saturation occurs at just **10% of full sunlight** (so option 1 is false). At low light conditions, **neither** C_3 nor C_4 plants respond to high CO_2 conditions (option 2 is false). At high light intensities, C_4 plants saturate at about $360\ \mu L\ L^{-1}$, whereas C_3 plants saturate only beyond $450\ \mu L\ L^{-1}$ (option 3 is true). Tomatoes and bell pepper are C_3 **plants**, not C_4 plants (option 4 is false).

10. Common NEET UG Pitfalls & Rapid Revision Checklist

High-Yield Traps to Avoid in the Exam

- **Trap 1 — Numbering vs. Sequence of Photosystems:** PS I and PS II are named in the order of their *discovery*, NOT their operational sequence. In the Z-scheme, **PS II (P680) functions first**, followed by **PS I (P700)**.
- **Trap 2 — Site of Proton Accumulation:** In chloroplasts, protons accumulate in the **thylakoid lumen** (resulting in a decrease in lumen pH), whereas in mitochondria during respiration, protons accumulate in the *intermembrane space*. Also remember that CF_1 protrudes into the **stroma**.
- **Trap 3 — RuBisCO & Calvin Cycle in C_4 Plants:** Never say C_4 plants lack the Calvin cycle! The Calvin cycle operates in **all photosynthetic plants**. In C_4 plants, it simply takes place exclusively inside the **bundle sheath cells** instead of mesophyll cells.
- **Trap 4 — Carbon Count of Acceptors vs. First Products:**
  - C_3 plant: Primary acceptor = **RuBP (5C)**; First product = **PGA (3C)**.
  - C_4 plant: Primary acceptor = **PEP (3C)**; First product = **OAA (4C)**.
- **Trap 5 — Water as a Limiting Factor:** Water stress limits photosynthesis primarily *indirectly* (by causing stomatal closure and leaf wilting) rather than by limiting the stoichiometric supply of H_2O for photolysis.

11. Concise Final Recap

- **Overall Equation:** $6CO_2 + 12H_2O \xrightarrow{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2$ (O_2 comes entirely from H_2O).
- **Pigments:** Chlorophyll *a* (chief pigment, blue-green; absorption peaks in blue and red), chlorophyll *b* (yellow-green), xanthophylls (yellow), carotenoids (yellow to yellow-orange).

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- **Light Reaction (Thylakoids):** Non-cyclic photophosphorylation (Grana; PS II P680 $\rightarrow$ ETS $\rightarrow$ PS I P700 $\rightarrow$ $\mathrm{NADP^{+}}$; yields ATP, $\mathrm{NADPH} + \mathrm{H^{+}}$, and $\mathrm{O_2}$). Cyclic photophosphorylation (Stroma lamellae; PS I only; wavelengths $> 680\,\mathrm{nm}$; yields ATP only).
- **Chemiosmosis:** Proton gradient builds up in thylakoid lumen (low pH) via water splitting, PQ proton pumping, and stromal $\mathrm{H^{+}}$ removal by NADP reductase; proton diffusion through $\mathrm{CF_0}$–$\mathrm{CF_1}$ ATP synthase drives ATP synthesis in the stroma.
- **Calvin Cycle (Stroma):** Carboxylation (RuBisCO) $\rightarrow$ Reduction ($2\,\mathrm{ATP} + 2\,\mathrm{NADPH}$ per $\mathrm{CO_2}$) $\rightarrow$ Regeneration ($1\,\mathrm{ATP}$ per $\mathrm{CO_2}$). Total per glucose ($6\,\mathrm{CO_2}$): $18\,\mathrm{ATP}$ and $12\,\mathrm{NADPH}$.
- $\mathrm{C_4}$ **& Photorespiration:** $\mathrm{C_4}$ plants (maize, sorghum) have Kranz anatomy, fix $\mathrm{CO_2}$ in mesophyll via PEPcase into OAA (4C), decarboxylate $\mathrm{C_4}$ acids in bundle sheath cells to concentrate $\mathrm{CO_2}$ around RuBisCO, eliminate photorespiration (which in $\mathrm{C_3}$ plants forms 3C phosphoglycerate + 2C phosphoglycolate), saturate at $360\,\mu\mathrm{L\,L^{-1}}\,\mathrm{CO_2}$, and thrive at 30–$40\,^{\circ}\mathrm{C}$.

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